

Ultrametric Quantum Error Correction

Testing Discrete Geometry in the Redundancy–Fidelity Tradeoff

CWI Summer School on Quantum Algorithms & Quantum Error Correction 24–28 August 2026
Poster Session: Wed 26 Aug, 16:30–18:00 · Poster Award Ceremony: Thu 27 Aug, 08:50

Panel 1 — The Hypothesis

- Standard view:** QEC assumes noise accumulates over a continuous topology; logical fidelity improves smoothly as redundancy is added.
- Alternative to test:** error propagation follows a discrete, hierarchical geometry — a Bruhat–Tits tree T_p — and redundancy quantizes in units of $(p+1)$.
- Signature to look for:** a staircase — flat plateaus followed by sudden jumps — instead of a smooth curve.

STATUS: HYPOTHESIS ONLY — NO EXPERIMENTAL RESULT CLAIMED

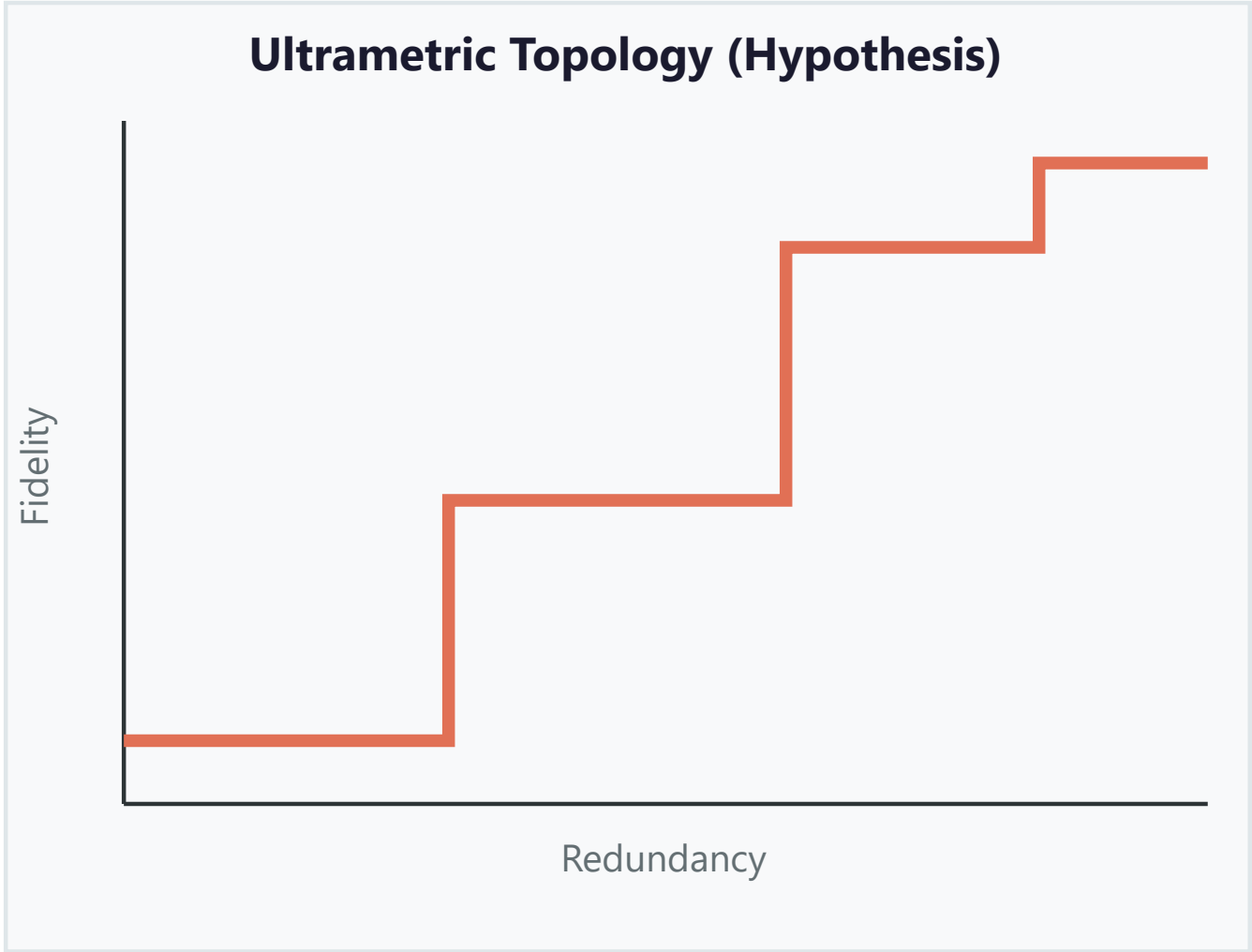
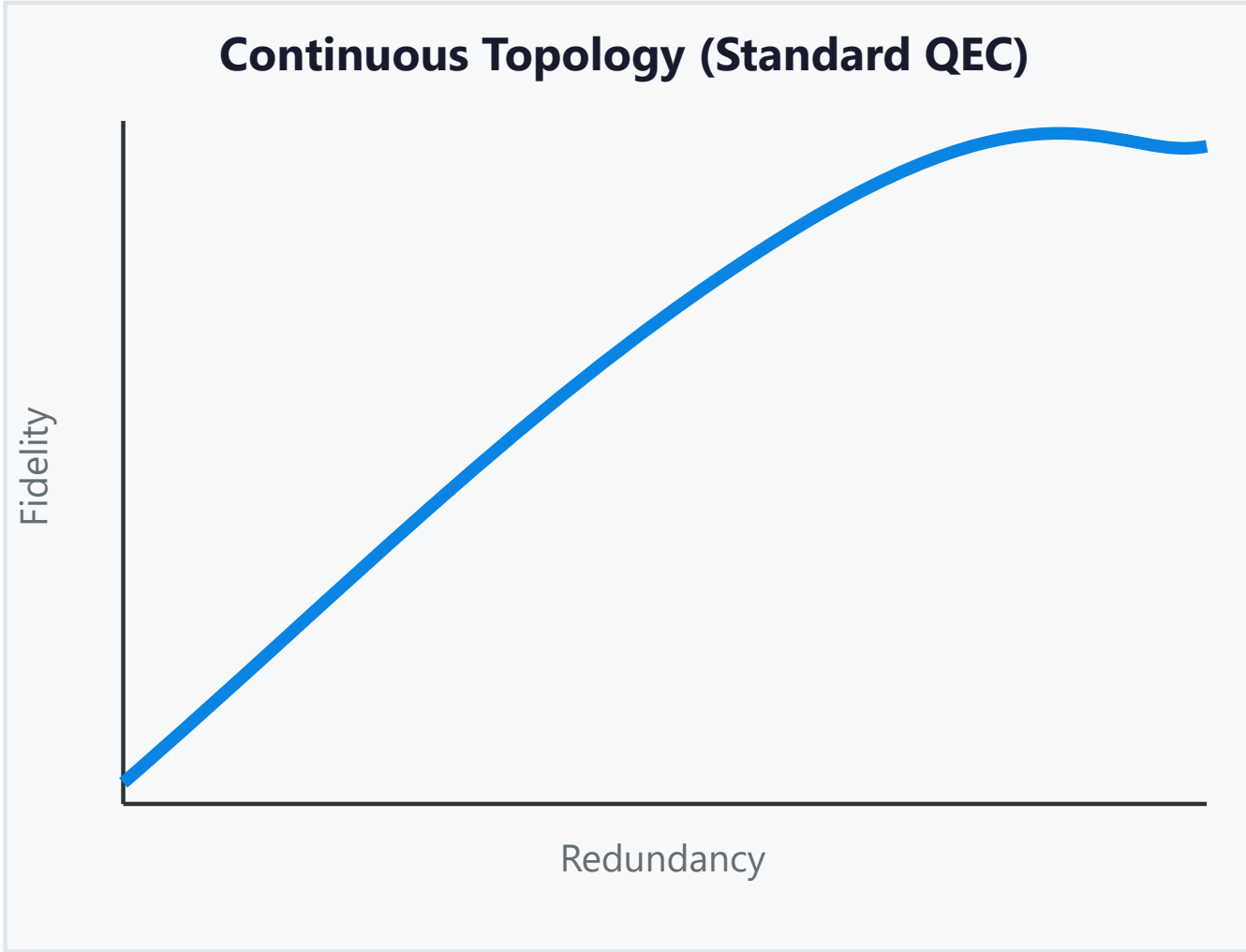
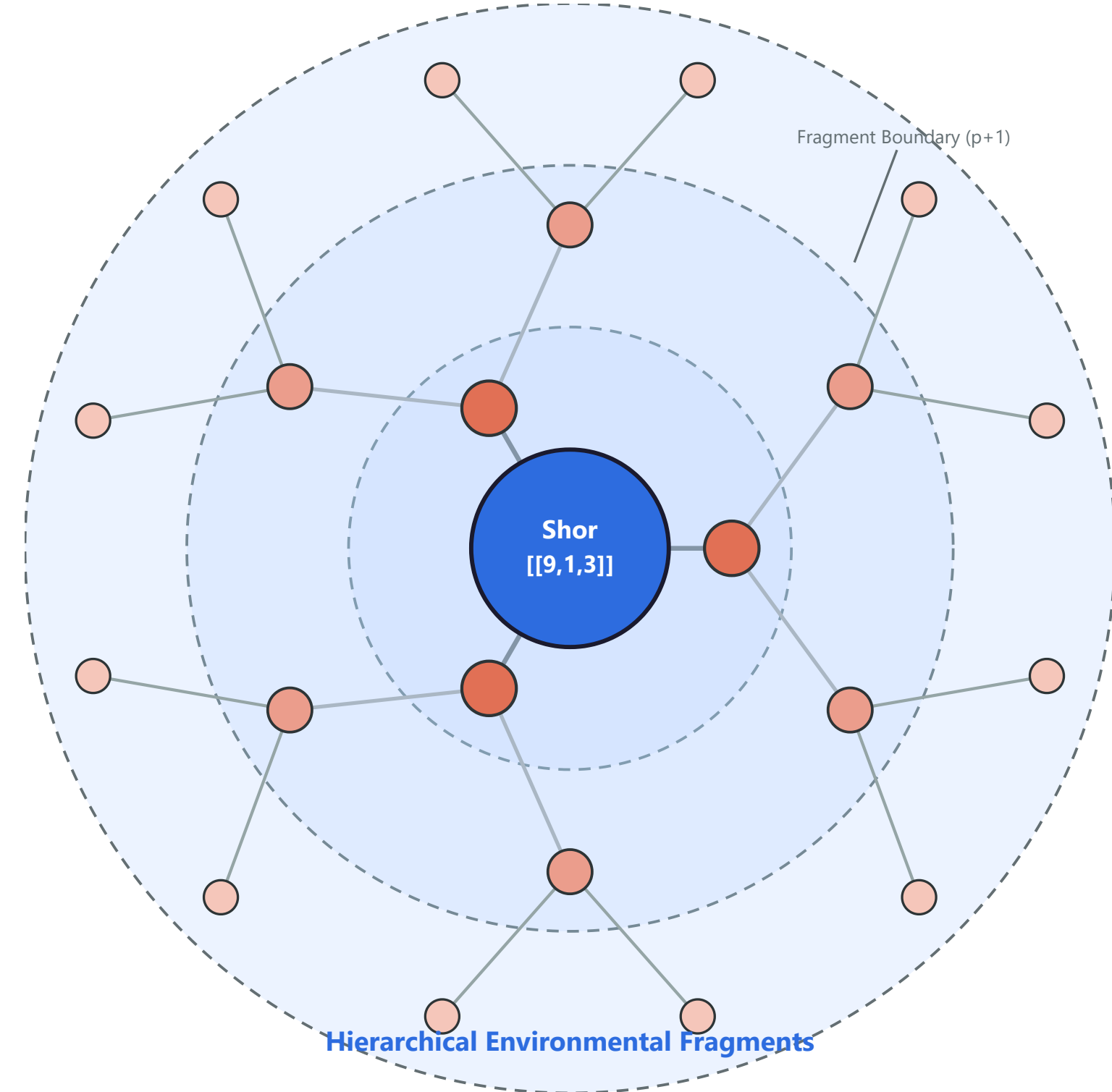


Figure 1: smooth fidelity curve (continuous topology, left) vs. staircase (ultrametric hypothesis, right). Hypothesis only; not observed.

Panel 2 — A Proposed Experiment (not yet run)



- Encode:** one logical qubit in the Shor $[[9,1,3]]$ code.
- Interact:** expose it to a fixed, structured environment of spectator qubits.
- Correct:** run the full error-correction circuit; measure final logical fidelity.
- Scale:** group environment qubits into larger hierarchical fragments; track fidelity across fragment boundaries.

Smooth shift with fragment size = continuous model. Staircase = ultrametric model.

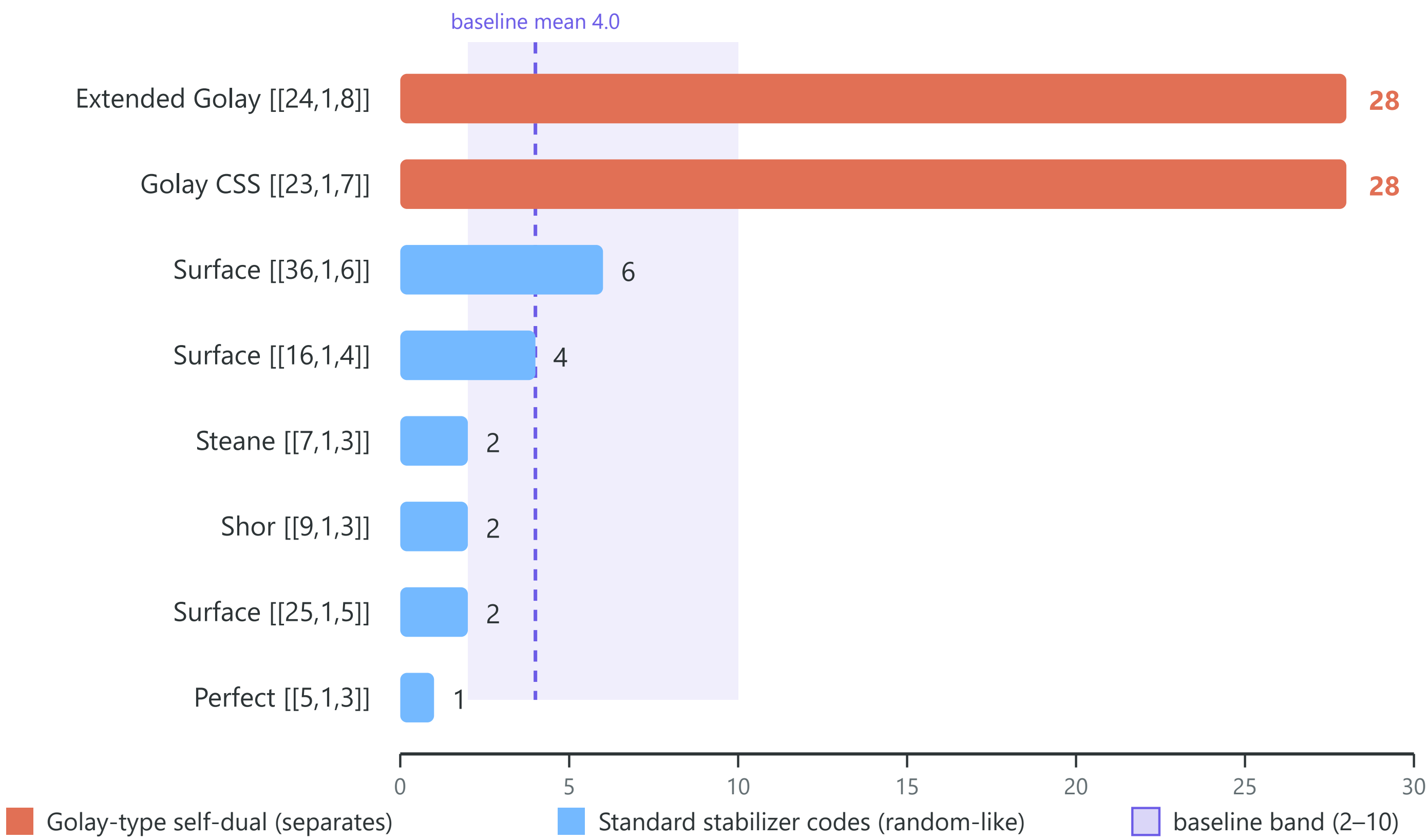
What exists today: a provisional numerical toy model (Archimedean Shadows) suggested redundancy may quantize in discrete steps (earlier versions reported 3, 6, 12, 24) — reported provisionally — and its own most seductive signature (the “forbidden window”) **failed its own falsification test**. Null reported, not hidden.

Panel 3 — The Cross-Check I Already Ran (it mostly failed — honestly)

Witness idea: 2-adic valuations as a structural fingerprint. Conjecture C7.3: the Mahler-spectrum valuation v_p^{max} separates “optimal” codes (28) from random codes (≈ 4).

v_p^{max} by code family — only Golay-type self-dual codes separate

Random-ensemble baseline (100 trials, $n=16$, $k=4$): mean 4.0, range 2–10 · Source: ACRP-06, 10.5281/zenodo.21737221



Code family	v_p^{max}
Golay CSS $[[23,1,7]]$	28
Extended Golay $[[24,1,8]]$	28
Perfect $[[5,1,3]]$	1
Steane $[[7,1,3]]$	2
Shor $[[9,1,3]]$	2
Surface $[[16,1,4]]$ / $[[25,1,5]]$ / $[[36,1,6]]$	4 / 2 / 6
Random ensemble (100 trials, $n=16$, $k=4$)	mean 4.0, range 2–10

Result: the conjecture does NOT generalize. Shor, Steane, perfect and surface codes are indistinguishable from the random baseline. The 28-vs-4 gap is specific to Golay-type self-dual symmetry — not a general “optimality” witness.

What it teaches: the first witness mostly failed, and the failure is informative — structure lives in an extreme-symmetry corner. **Open:** is there a refined witness (valuations of the full weight distribution $v_2(A_i)$, not just the Mahler maximum) that discriminates where v_p^{max} fails — and does any witness connect to the staircase?

Panel 4 — Value & Open Questions

A smooth curve validates continuous models; a staircase would force code architectures designed for hierarchical noise.

- What physical mechanism would produce tree-like error geometry?
- Does any valuation witness predict where the staircase should be most visible? (v_p^{max} does not.)
- How would discrete geometry alter asymptotic fault-tolerance thresholds?

I'm here to learn: has anyone in this room seen staircase-like scaling — or a reason it must be wrong?